

RAG Pipeline Evaluation Report

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Frameworks: DeepEval · Ragas · LLM-as-Judge

Project	Insurance Claims Intelligence Assistant
Dataset	fraud_oracle.csv — 15,420 vehicle insurance records
LLM	gpt-4o-mini
Embedding	text-embedding-3-small
Test Cases	3

1. Test Cases

Test Case 1: High Fraud — Classic Pattern

Query	Sedan collision policy holder fault urban area police report not filed no witness more than 4 past claims
Response	The claim is invalid due to the absence of a police report, the policyholder being at fault, and having more than 4 past cl
Expected	High fraud risk with multiple indicators — no police report, no witness, policy holder fault, and excessive past claims. Es
Context	• Policy Number: 10874. Policy Type: Sedan - Collision. Base Policy: Collision. Fraud Found: Y. Accident Area: Urban. F • Policy Number: 15294. Policy Type: Sedan - Collision. Base Policy: Collision. Fraud Found: N. Accident Area: Urban. F • Policy Number: 2940. Policy Type: Sedan - Collision. Base Policy: Collision. Fraud Found: N. Accident Area: Urban. F

Test Case 2: Low Fraud — Legitimate Claim

Query	Sedan collision third party fault police report filed witness present no previous claims
Response	The claim is valid as it includes a filed police report, indicates third party fault, and shows no previous claims. The fraud
Expected	Low to medium fraud risk. Third party fault with police report and witness are positive indicators. Minor investigation may
Context	• Policy Number: 3381. Policy Type: Sedan - Collision. Base Policy: Collision. Fraud Found: N. Accident Area: Urban. F • Policy Number: 13315. Policy Type: Sedan - Collision. Base Policy: Collision. Fraud Found: N. Accident Area: Urban. F • Policy Number: 8019. Policy Type: Sedan - Collision. Base Policy: Collision. Fraud Found: N. Accident Area: Urban. F

Test Case 3: Medium Fraud — Mixed Signals

Query	Sport vehicle collision policy holder fault urban area no police report 2 to 4 past claims witness present
Response	The claim is invalid due to the policyholder being at fault and the absence of a police report. High fraud probability (0.7)
Expected	Medium fraud risk. No police report is a concern but witness present and moderate past claims create mixed signals. In
Context	• Policy Number: 4717. Policy Type: Sport - Collision. Base Policy: Collision. Fraud Found: Y. Accident Area: Urban. Fa • Policy Number: 11331. Policy Type: Sport - Collision. Base Policy: Collision. Fraud Found: Y. Accident Area: Urban. F • Policy Number: 13856. Policy Type: Sport - Collision. Base Policy: Collision. Fraud Found: Y. Accident Area: Urban. F

2. DeepEval Results

DeepEval measures Faithfulness (≥ 0.7), Answer Relevancy (≥ 0.7), and Contextual Precision (≥ 0.6) — checking if the AI response is grounded in retrieved context and relevant to the query.

Test Case	Faithfulness (≥ 0.70)	Answer Relevancy (≥ 0.70)	Contextual Precision (≥ 0.60)	Overall
High Fraud — Classic Pattern	0.300	1.000	0.600	FAIL ✗
Low Fraud — Legitimate Claim	0.300	1.000	0.500	FAIL ✗
Medium Fraud — Mixed Signals	0.300	1.000	0.500	FAIL ✗
Summary				0/3 PASSED

3. Ragas Results

Ragas measures Faithfulness, Answer Relevancy, and Context Recall — evaluating whether retrieved context is sufficient to generate a correct answer.

Metric	Average Score	Threshold	Status
Faithfulness	0.600	≥ 0.50	PASS ✓
Answer Relevancy	0.933	≥ 0.50	PASS ✓
Context Recall	0.367	≥ 0.50	FAIL ✗

4. LLM-as-Judge Results

A second GPT-4o-mini instance independently scores each response on Faithfulness, Helpfulness, and Accuracy (1–5 scale).

Test Case	Faithfulness (1–5)	Helpfulness (1–5)	Accuracy (1–5)	Average	Reasoning
High Fraud — Classic Pattern	5.0/5	5.0/5	5.0/5	5.0	The overall quality of this answer is high, as it clearly identifies the reasons for the claim's invalidity and outlines...
Low Fraud — Legitimate Claim	2.0/5	3.0/5	2.0/5	2.33	The overall quality of this answer is poor, as it incorrectly states that a police report was filed when the context ind...
Medium Fraud — Mixed Signals	4.0/5	5.0/5	3.0/5	4.0	The overall quality of this answer is high, as it effectively identifies key risk factors, provides a clear rationale fo...

5. Overall Summary

Framework	Result	Status
DeepEval	0/3 test cases passed	FAIL ✗
Ragas	Average score: 0.633	FAIL ✗
LLM-as-Judge	Average score: 3.78/5.00	PASS ✓
OVERALL	System evaluation complete	NEEDS IMPROVEMENT

The ClaimsIQ RAG pipeline has been evaluated across three independent frameworks. DeepEval passed 0 of 3 test cases on faithfulness, answer relevancy, and contextual precision. Ragas achieved an average score of 0.633 across faithfulness, answer relevancy, and context recall metrics. LLM-as-Judge averaged 3.78/5.00 on faithfulness, helpfulness, and accuracy. Some metrics are below threshold — prompt tuning or retrieval improvements recommended.